

output: one value

output: a sequence

**input:
one value**

one-to-one

image $\rightarrow$ cat or not
an ordinary ANN or CNN,
not an RNN

one-to-many

image $\rightarrow$ caption
input once, output
at every step

**input:
a sequence**

many-to-one

review $\rightarrow$ sentiment
output after
the last step

many-to-many

same length: word $\rightarrow$ tag
different lengths:
English $\rightarrow$ Hindi
(sequence-to-sequence)